Quiz 1

Name: _____

Show your work. Unsupported answers may receive limited credit. **Total: 10 points**

1. Information from a formula. **[7 points]**

Let

$$h(x) = \frac{x^2 - 4}{x + 1}.$$

(a) State the domain in interval notation. [1 point]

(b) Find all x-intercepts. Write each intercept as an ordered pair. [2 points]

(c) Find the y-intercept. Write it as an ordered pair. [1 point]

(d) Evaluate $h(-2)$. [1 point]

(e) Determine whether $(3, 0)$ lies on the graph of h. Justify your answer. [2 points]

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2. Even or odd? **[3 points]**

Let

$$g(x) = x^3 - 4x.$$

Determine whether g is **even**, **odd**, or **neither**. Show your calculation of $g(-x)$ and compare it with $g(x)$ and $-g(x)$.